

Permutation crossword grids with one or two complete rook placements

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Abstract

For a permutation w , let $r(w)$ count the complete nonattacking rook placements on the square crossword grid whose black cells are (i, w_i) . We prove that every permutation containing 2143 or 3412 has at least three such placements. Together with the skew-merged implication of Lewis and Won, this proves the equivalence stated in Conjecture 3.9 of their arXiv version 1: $r(w) \in \{1, 2\}$ if and only if w is skew-merged. The proof uses injections for four extremal deletions, reduces the terminal cases to a two-parameter family up to grid symmetry, and constructs three placements uniformly in that family.

1 Statement and context

For $n \geq 1$ and $w = w_1 \cdots w_n \in S_n$, the grid $\text{Grid}(w)$ has n rows and n columns. Its black cells are exactly (i, w_i) ; all other cells are white. Rows are numbered from top to bottom and columns from left to right. An *across word* is a nonempty maximal contiguous white run in a row; a *down word* is the corresponding run in a column. Runs of length one are included. A *complete nonattacking rook placement* is a set of white cells meeting every across and every down word exactly once. Black cells block attacks, so two rooks may occupy the same geometric row or column if separated by its black cell. Write $r(w)$ for the number of complete placements. There are $2n - 2$ words in each direction, hence $2n - 2$ rooks in each complete placement.

A permutation is *skew-merged* if its entries can be partitioned into an increasing subsequence and a decreasing subsequence, either possibly empty. Containment of a pattern means containment as an order-isomorphic subsequence, and $\text{Av}(2143, 3412)$ denotes simultaneous avoidance of those two patterns. We use the classical characterization of skew-merged permutations as $\text{Av}(2143, 3412)$, due to Stankova [2, Theorem 2.9], in the form quoted by Lewis and Won [1, p. 14].

Theorem 1. *For every $n \geq 1$ and every $w \in S_n$,*

$$r(w) \in \{1, 2\} \iff w \text{ is skew-merged.}$$

In particular, if w contains 2143 or 3412, then $r(w) \geq 3$. With the convention that the empty grid has one empty placement, the equivalence also holds for $n = 0$.

Lewis and Won [1, Proposition 3.8] prove that a skew-merged permutation has one or two placements. Their Conjecture 3.9 asks for the converse. Their Proposition 3.3 and Remark 3.4 also give an injection under deletion of the maximum entry. We recall that injection explicitly and use its symmetric variants. The additional argument is a necessary classification of the permutations for which none of these four deletions retains a forbidden pattern, followed by explicit constructions

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in all remaining cases. Only extremal deletion is used; no monotonicity under arbitrary pattern deletion is assumed.

All comparisons with the conjecture refer to the inspected version 1 of [1]. Current versions, subsequent literature and priority have not been established here, and no priority claim is made. As broader context for permutation enumeration supplemented by exact finite checks, see Zeng [3]; that reference concerns cycle lengths and is not used in the proof of Theorem 1.

2 Extremal deletion

Reflections in either grid axis and transposition biject white words and complete placements. The induced operations on permutations preserve $\text{Av}(2143, 3412)$, since they permute the two forbidden patterns. Deleting an entry always includes standardizing the remaining values.

Lemma 2. *If v is obtained from w by deleting its maximum, minimum, first or last entry, then $r(v) \leq r(w)$.*

Proof. We give the maximum-deletion injection from [1, Proposition 3.3 and Remark 3.4], including its recovery map. Let the maximum n occur in row i . Removing row i and column n gives $\text{Grid}(v)$. For $n = 1$, both grids have their unique empty placement. Suppose $n \geq 2$, and let j be the row containing $n - 1$. Lift a complete placement R on $\text{Grid}(v)$ to the undeleted rows and columns of $\text{Grid}(w)$.

If $i = n$, add rooks at

$$(n, w_{n-1}) \quad \text{and} \quad (j, n).$$

The new across words are the west word of row n and the east word of row j . The new down words are the south word of column w_{n-1} and the north word of column n . The added rooks fill exactly these four words, and all inherited words retain their rooks. Removing the added rooks recovers R . The case $i = 1$ is its row reflection.

For $1 < i < n$, reflect rows if needed so that $j < i$. In the lifted placement, row $i + 1$ has a unique rook $(i + 1, t)$ east of its black cell: its black value is at most $n - 2$, since $n - 1$ occurs in row j . Remove $(i + 1, t)$ and add

$$(i, t), \quad (i + 1, n), \quad (j, n). \tag{1}$$

The cells (i, t) and $(i + 1, t)$ lie in the same down word: they are adjacent and neither is black. The first new rook therefore preserves that down word and fills the new west word of row i . The second fills the vacated east word of row $i + 1$ and the new south word of column n . The third fills the new east word of row j and the new north word of column n . All other words remain covered once.

The resulting placement has exactly one rook in row i , whose column recovers t . Remove it and the two column- n rooks, restore $(i + 1, t)$, and delete row i and column n . This recovers the input, proving injectivity. Grid symmetries carry maximum deletion to each of the other three deletions in the statement. $\square$

3 The terminal family

Call a permutation *terminal* if it contains 2143 or 3412, but deleting any of its first, last, minimum or maximum entries leaves a permutation avoiding both patterns.

Lemma 3. *Every terminal permutation is, up to grid symmetry, of the form*

$$C(p, q) = (2, n-1, n-2, \dots, q+4, 1, 3, 4, \dots, q+2, n, q+3), \quad (2)$$

where $p, q \geq 0$, $n = p + q + 4$, the descending block has p entries, and the increasing block has q entries. Empty blocks are omitted.

Proof. Every occurrence of a forbidden pattern must contain the first, last, minimum and maximum entries of the permutation: an occurrence omitting one would survive its deletion. Complement values if necessary so that a chosen forbidden occurrence is 2143. Its first and last positions must be the global first and last positions, and its least and greatest values must be the global minimum and maximum. These four extremal entries thus occupy four distinct roles. In position order write them as

$$A = a, \quad B = 1, \quad C = n, \quad D = d, \quad 1 < a < d < n.$$

Let X, Y, Z be the position blocks between A, B , between B, C , and between C, D , respectively.

If $x \in X$ satisfied $x < d$, then (x, B, C, D) would be a 2143 occurrence omitting A . Thus $x > d$. If $y \in Y$ satisfied $y < a$, then (A, y, C, D) would be 2143; if $y > d$, then (A, B, y, D) would be 2143. Hence $a < y < d$. If $z \in Z$ satisfied $z > a$, then (A, B, C, z) would be 2143, so $z < a$. Equality is excluded because all entries are distinct.

An ascent $x < x'$ in X gives the occurrence (x, x', B, D) of 3412. A descent $y > y'$ in Y gives (A, B, y, y') of 2143. An ascent $z < z'$ in Z gives (A, C, z, z') of 3412. Each occurrence omits a required extremal entry. Consequently X decreases, Y increases, and Z decreases. Moreover, X and Z cannot both be nonempty: for $x \in X$ and $z \in Z$, the subsequence (A, x, B, z) is 3412, because $1 < z < a < x$.

A half-turn, namely reversal followed by complementation, preserves 2143 and exchanges the roles of X and Z . We may take Z empty. Any value strictly between 1 and a would have to be in Z , so $a = 2$. Put $p = |X|$ and $q = |Y|$. The forced value ranges and orders then give $d = q + 3$ and exactly (2). $\square$

Only the necessary form in Lemma 3 is needed; no converse classification is used.

4 Three placements in each terminal form

A *rectangular switch* replaces rooks $(i, j), (k, \ell)$ by $(i, \ell), (k, j)$, provided the two columns lie in the same across word in each indicated row and the two rows lie in the same down word in each indicated column. It preserves the occupied words and hence completeness and nonattack. We exhibit a base placement and two legal switches, each applied separately, in every $C(p, q)$. Throughout, an integer interval whose lower endpoint exceeds its upper endpoint is empty.

Lemma 4. *For every $p, q \geq 0$, $r(C(p, q)) \geq 3$.*

Proof. There are three cases.

Case 1: $p = 0$. Here $n = q + 4$ and $w = (2, 1, 3, 4, \dots, n-2, n, n-1)$. Set

$$R = \{(1, 1), (n, n)\} \cup \{(j-2, j) : 3 \leq j \leq n\} \cup \{(j+2, j) : 1 \leq j \leq n-2\}. \quad (3)$$

These are $2n - 2$ distinct white cells. In the north words, column 1 receives $(1, 1)$, column 2 has no such word, and columns 3 through n receive $(j-2, j)$. In the south words, columns 1 through $n-2$ receive $(j+2, j)$, column $n-1$ has no such word, and column n receives (n, n) . Rows 1 and n

receive both their west and east rooks; row 2 has its sole east rook and row $n - 1$ its sole west rook. Each remaining row i , with $3 \leq i \leq n - 2$, has rooks in columns $i - 2$ and $i + 2$, on opposite sides of its black cell. This also verifies the case $n = 4$.

One switch uses $(1, 3), (2, 4)$. Both columns lie east of both row black cells, and their black cells lie below row 2. A second switch uses $(n - 1, n - 3), (n, n - 2)$. Both columns lie west of both row black cells, and their black cells lie above row $n - 1$. Thus both switches are legal. Their row pairs are disjoint, so applying either one separately gives two different placements, both distinct from R .

Common notation for $p \geq 1$. Put

$$u = p + 1, \quad b = p + 2, \quad c = n - 1, \quad d = q + 3.$$

The descending black block is in rows $k = 2, \dots, p + 1$, with black value $n + 1 - k$. Rows b, c, n have black values $1, n, d$, respectively. Define

$$F = \{(1, 1), (n, n)\} \cup \bigcup_{k=2}^{p+1} \{(k - 1, n + 1 - k), (k, n + 2 - k)\}. \quad (4)$$

The $2p + 2$ cells of F are distinct and white. They fill both words in row 1 and rows $2, \dots, p$, the east word of row u , and the east word of row n . In columns, they fill the north word of column 1, the north words of columns $d + 1, \dots, n - 1$, the south words of columns $d + 2, \dots, n - 1$, and both words of column n . These descriptions include $p = 1$, where some intervals are empty.

Case 2: $p \geq 1, q = 0$. Here $d = 3, n = p + 4$, and we take

$$R = F \cup \{(u, 2), (b, 3), (c, 4), (n, 1)\}. \quad (5)$$

The added rooks fill, in order, the west word of row u , the east word of row b , the west word of row c , and the west word of row n . They also fill exactly the four remaining column words: south of column 2, north of column 3, south of column 4, and south of column 1. Thus R is a complete placement with $2n - 2$ rooks.

Switch $(b, 3), (c, 4)$ to $(b, 4), (c, 3)$. The black values in rows b, c are $1, n$, with $n \geq 5$; the black cells of columns 3, 4 are in rows n, u , with $u < b < c < n$. The same four words remain occupied. Alternatively, switch $(u, 2), (b, 3)$ to $(u, 3), (b, 2)$. The black values in rows u, b are $4, 1$, and the black rows of columns 2, 3 are $1, n$, so this switch is legal as well. The first leaves row u unchanged while the second changes it, proving that the three placements are distinct, including when $p = 1$.

Case 3: $p \geq 1, q \geq 1$. The central increasing black entries have value $t + 2$ in row $p + t + 2$, for $1 \leq t \leq q$. Set

$$\begin{aligned} R = F \cup \{ & (u, 2), (b, 3), (c, q + 4), (n, q + 2), (p + 3, 1) \} \\ & \cup \{(p + t + 2, t + 1) : 2 \leq t \leq q\} \\ & \cup \{(p + t + 2, t + 3) : 1 \leq t \leq q\}. \end{aligned} \quad (6)$$

There are $2n - 2$ distinct white cells. The added rooks give row u its west rook, row b its east rook, and rows c, n their west rooks. Each central row receives a west rook (column 1 in the first such row, and black value minus 1 thereafter) and an east rook at black value plus 1. Along with F , this fills every across word exactly once.

For the down words, the inverse-permutation positions are

$$\begin{aligned} \text{pos}(1) &= p + 2, & \text{pos}(2) &= 1, & \text{pos}(j) &= p + j & (3 \leq j \leq q + 2), \\ \text{pos}(d) &= n, & \text{pos}(j) &= n + 1 - j & (d + 1 \leq j \leq n - 1), & \text{pos}(n) &= n - 1. \end{aligned}$$

The words not filled by F and their rook rows are exactly as follows:

Column(s)	Word	Rook row in R
1	south	$p + 3$
2	south	u
$3 \leq j \leq d$	north	$p + j - 1$
$3 \leq j \leq q + 1$	south	$p + j + 1$
$q + 2$	south	n
$q + 4 = d + 1$	south	c

The displayed positions show that each rook lies on the stated side of its black cell. The south interval $3 \leq j \leq q + 1$ is empty when $q = 1$ and never overlaps column $q + 2$. Column d has no south word. Thus every down word is filled once.

The first switch uses $(n - 2, q + 3), (n - 1, q + 4)$. The black values in these rows are $q + 2, n$, so both candidate columns stay in the east word of row $n - 2$ and the west word of row $n - 1$. The black rows of columns $q + 3, q + 4$ are n, u , respectively, with $u < n - 2$, so their down words are unchanged. The second switch uses $(u, 2), (b, 3)$. The row black values are $q + 4, 1$, and the column black rows are $1, p + 3$, with $p + 3 > b$. It too preserves all four word identities. The row pairs (u, b) and $(n - 2, n - 1)$ are disjoint even at $q = 1$, so the base and the two separately switched placements are distinct. $\square$

5 Completion of the proof and verification scope

Proof of Theorem 1. We prove by strong induction on length that every permutation containing 2143 or 3412 has at least three placements. Below length four there is nothing to prove. For such a permutation w , if an extremal deletion retains a forbidden occurrence, the smaller grid has at least three placements by induction, and Lemma 2 injects them into $\text{Grid}(w)$. Otherwise w is terminal. Lemma 3 takes it by grid symmetry to some $C(p, q)$, and Lemma 4 gives three placements there. The inverse symmetry gives three distinct placements on $\text{Grid}(w)$.

The characterization of skew-merged permutations as $\text{Av}(2143, 3412)$ now proves that $r(w) \in \{1, 2\}$ implies that w is skew-merged. The other implication is [1, Proposition 3.8]. When $n = 1$, there are no white words and the empty placement is unique. The empty-permutation convention gives the same conclusion at $n = 0$. $\square$

The proof is uniform in the parameters and does not depend on finite enumeration. Exact computations supplied supporting checks: the formula placements were checked on all 961 pairs $0 \leq p, q \leq 30$, and a recursive implementation constructed three distinct placements for all 39,182 non-skew-merged permutations through length eight. A separately implemented word-scanning verifier checked all 117,546 resulting placements. It also checked the formulas on 967 parameter pairs (the same box and six additional pairs, with maximum length 208), and checked validity and injectivity of the maximum-deletion map on all 38,129 input placements arising from all 5,913 permutations of lengths one through seven. These ranges describe finite checks, not an exhaustive verification at arbitrary length. The argument asserts at least three placements in the terminal family; it does not enumerate all of them.

Computational and AI assistance. AI assistance was used in developing the argument, writing exact-check programs, reviewing the proof in a separate model context, and preparing this manuscript. The recursive-construction replay reused the original constructor with a separately written verifier; it was not a second independent implementation of the constructor. A separate model review is not external peer review, statistical independence, or formal proof verification. The universal result rests on the mathematical argument above and the stated prior results. The original article of Stankova was not directly inspected; its characterization is used as quoted in the inspected Lewis–Won source.

References

- [1] Joel Brewster Lewis and Robert Won, *Non-attacking rook placements on crossword grids*, arXiv:2609.03081v1, 2 September 2026. <https://arxiv.org/abs/2609.03081v1>.
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